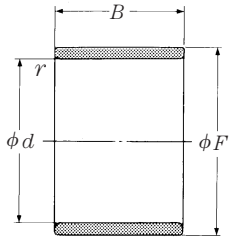
FIR

Bore Diameter 55 — 70 mm

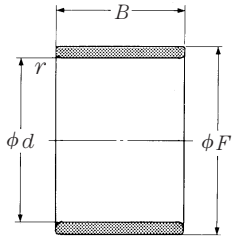


Boundary Dimensions (mm)				Inner Ring Numbers	Mass (kg)
d	F	B	r min		approx
55	60	25	0.3	**FIR-556025-1**	0.088
	60	35	0.3	**FIR-556035-1**	0.123
	63	25	1	**FIR-556325-1**	0.145
	63	26	1	**FIR-556326-1**	0.150
	63	34	1	**FIR-556334-1**	0.197
	63	45	1	**FIR-556345-1**	0.260
	63	46	1	**FIR-556346-1**	0.266
	65	30	1	**FIR-556530-1**	0.221
	65	60	1	**FIR-556560-1**	0.441
60	68	25	1	**FIR-606825-1A**	0.157
	68	26	1	**FIR-606826-1**	0.163
	68	34	1	**FIR-606834-1**	0.213
	68	35	0.6	**FIR-606835-1**	0.220
	68	45	1	**FIR-606845-1**	0.282
	68	46	1	**FIR-606846-1**	0.289
	70	30	1	**FIR-607030-1**	0.239
	70	60	1	**FIR-607060-1**	0.478
65	72	25	1	**FIR-657225-1**	0.147
	72	26	1	**FIR-657226-1**	0.153
	72	34	1	**FIR-657234-1**	0.200
	72	45	1	**FIR-657245-1**	0.264
	72	46	1	**FIR-657246-1**	0.270
	73	35	1	**FIR-657335-1**	0.237
	75	30	1	**FIR-657530-1**	0.257
	75	60	1	**FIR-657560-1**	0.515
70	80	25	1	**FIR-708025-1**	0.230
	80	30	1	**FIR-708030-1**	0.276
	80	31	1	**FIR-708031-1**	0.285
	80	35	1	**FIR-708035-1**	0.322
	80	40	1	**FIR-708040-1**	0.368
	80	54	1	**FIR-708054-1**	0.496
	80	55	1	**FIR-708055-1**	0.505
	80	60	1	**FIR-708060-1**	0.551

Matching Solid Needle Roller Bearings			
RLM	RNA	RNAF	RNAFW
RLM607225			
RLM607235-1			
	RNA4911		
	RNA4911TT		
	RNA5911		
	RNA6911		
	RNA6911TT		
		RNAF658530	
			RNAFW658560
RLM688225-1	RNA4912		
	RNA4912TT		
	RNA5912		
RLM688235-1			
	RNA6912		
	RNA6912TT		
		RNAF709030	
			RNAFW709060
	RNA4913		
	RNA4913TT		
	RNA5913		
	RNA6913		
	RNA6913TT		
RLM739035-1			
		RNAF759530	
			RNAFW759560
RLM809525-1			
	RNA4914	RNAF8010030	
	RNA4914TT		
RLM809535-1			
	RNA5914		
	RNA6914		
	RNA6914TT		
			RNAFW8010060

FIR

Bore Diameter 75 — 100 mm

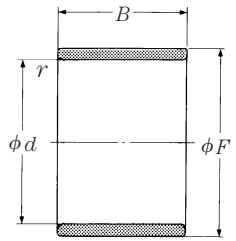


Boundary Dimensions (mm)				Inner Ring Numbers	Mass (kg)
d	F	B	r min		approx
75	85	25	1	**FIR-758525-1**	0.245
	85	30	1	**FIR-758530-1**	0.294
	85	31	1	**FIR-758531-1**	0.304
	85	35	1	**FIR-758535-1**	0.343
	85	40	1	**FIR-758540-1**	0.392
	85	54	1	**FIR-758554-1**	0.529
	85	55	1	**FIR-758555-1**	0.539
80	90	25	1	**FIR-809025-1**	0.260
	90	30	1	**FIR-809030-1**	0.312
	90	31	1	**FIR-809031-1**	0.323
	90	40	1	**FIR-809040-1**	0.417
	90	54	1	**FIR-809054-1**	0.562
	90	55	1	**FIR-809055-1**	0.573
85	95	26	1	**FIR-859526-1**	0.287
	95	30	1	**FIR-859530-1**	0.331
	100	35	1.1	**FIR-8510035-1**	0.595
	100	36	1.1	**FIR-8510036-1**	0.612
	100	46	1.1	**FIR-8510046-1**	0.782
	100	63	1.1	**FIR-8510063-1**	1.071
	100	64	1.1	**FIR-8510064-1**	1.088
90	100	26	1	**FIR-9010026-1**	0.303
	100	30	1	**FIR-9010030-1**	0.349
	105	35	1.1	**FIR-9010535-1**	0.627
	105	36	1.1	**FIR-9010536-1**	0.645
	105	46	1.1	**FIR-9010546-1**	0.824
	105	63	1.1	**FIR-9010563-1**	1.129
95	110	35	1.1	**FIR-9511035-1**	0.659
	110	36	1.1	**FIR-9511036-1**	0.678
	110	46	1.1	**FIR-9511046-1**	0.867
	110	63	1.1	**FIR-9511063-1**	1.187
	110	64	1.1	**FIR-9511064-1**	1.206
100	115	40	1.1	**FIR-10011540-1**	0.790
	115	41	1.1	**FIR-10011541-1**	0.810
	115	54	1.1	**FIR-10011554-1**	1.067

Matching Solid Needle Roller Bearings			
RLM	RNA	RNAF	RNAFW
RLM8510525-1			
	RNA4915	RNAF8510530	
	RNA4915TT		
RLM8510535-1			
	RNA5915		
	RNA6915		
	RNA6915TT		
RLM9011025-1			
	RNA4916	RNAF9011030	
	RNA4916TT		
	RNA5916		
	RNA6916		
	RNA6916TT		
RLM9511526-1			
		RNAF9511530	
	RNA4917		
	RNA4917TT		
	RNA5917		
	RNA6917		
	RNA6917TT		
RLM10012026-1			
		RNAF10012030	
	RNA4918		
	RNA4918TT		
	RNA5918		
	RNA6918		
	RNA4919		
	RNA4919TT		
	RNA5919		
	RNA6919		
	RNA6919TT		
	RNA4920		
	RNA4920TT		
	RNA5920		

FIR

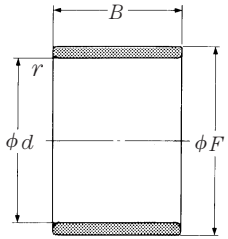
Bore Diameter 110 — 220 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (kg)	Matching Solid Needle Roller Bearings			
d	F	B	r min		approx	RLM	RNA	RNAF	RNAFW
110	120	30	1	**FIR-11012030-1**	0.423		RNA4822		
	125	40	1.1	**FIR-11012540-1**	0.864		RNA4922		
	125	41	1.1	**FIR-11012541-1**	0.885		RNA4922TT		
	125	54	1.1	**FIR-11012554-1**	1.166		RNA5922		
120	130	30	1	**FIR-12013030-1**	0.459		RNA4824		
	135	45	1.1	**FIR-12013545-1**	1.054		RNA4924		
	135	60	1.1	**FIR-12013560-1**	1.406		RNA5924		
130	145	35	1.1	**FIR-13014535-1**	0.884		RNA4826		
	150	50	1.5	**FIR-13015050-1**	1.715		RNA4926		
	150	67	1.5	**FIR-13015067-1**	2.298		RNA5926		
140	155	35	1.1	**FIR-14015535-1**	0.949		RNA4828		
	160	50	1.5	**FIR-14016050-1**	1.838		RNA4928		
	160	67	1.5	**FIR-14016067-1**	2.463		RNA5928		
150	165	40	1.1	**FIR-15016540-1**	1.158		RNA4830		
	170	60	2	**FIR-15017060-1**	2.352		RNA4930		
160	175	40	1.1	**FIR-16017540-1**	1.231		RNA4832		
	180	60	2	**FIR-16018060-1**	2.499		RNA4932		
170	185	45	1.1	**FIR-17018545-1**	1.468		RNA4834		
	190	60	2	**FIR-17019060-1**	2.646		RNA4934		
180	195	45	1.1	**FIR-18019545-1**	1.551		RNA4836		
	205	69	2	**FIR-18020569-1**	4.068		RNA4936		
190	210	50	1.5	**FIR-19021050-1**	2.450		RNA4838		
	215	69	2	**FIR-19021569-1**	4.280		RNA4938		
200	220	50	1.5	**FIR-20022050-1**	2.573		RNA4840		
	225	80	2.1	**FIR-20022580-1**	5.207		RNA4940		
220	240	50	1.5	**FIR-22024050-1**	2.818		RNA4844		
	245	80	2.1	**FIR-22024580-1**	5.697		RNA4944		

FIR

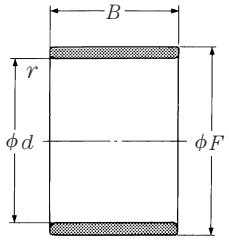
Bore Diameter 8 — 20 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (g)	Matching Solid Needle Roller Bearings	
						RLM	
d	F	B	r min		approx		
8	12	12.2	0.3	**FIR-81212-2**	5.9	RLM1212	RLM121912
	12	15.2	0.3	**FIR-81215-2**	7.4		RLM121915
10	14	16.2	0.3	**FIR-101416-2**	9.5	RLM1416	
	14	20.2	0.3	**FIR-101420-2**	11.9	RLM1420	
	15	8.2	0.3	**FIR-10158-2**	6.2	RLM158	
	15	12.2	0.3	**FIR-101512-2**	9.3		RLM152212
	15	15.2	0.3	**FIR-101515-2**	11.6	RLM1515	RLM152215
	15	20.2	0.3	**FIR-101520-2**	15.5	RLM1520	RLM152220
12	16	16.2	0.3	**FIR-121616-2**	11.1	RLM1616	
	16	20.2	0.3	**FIR-121620-2**	13.9	RLM1620	
	17	10.2	0.3	**FIR-121710-2**	9.0	RLM1710	
	17	15.2	0.5	**FIR-121715-2**	13.5		RLM172415
	17	20.2	0.5	**FIR-121720-2**	17.9	RLM1720	
	17	25.2	0.5	**FIR-121725-2**	22.4		RLM172425
15	18	15.2	0.5	**FIR-151815-2**	9.2	RLM1815	
	18	17.2	0.5	**FIR-151817-2**	10.4	RLM1817	
	18	20.2	0.5	**FIR-151820-2**	12.3	RLM1820	
	18	25.2	0.5	**FIR-151825-2**	15.3	RLM1825	
	19	16.2	0.5	**FIR-151916-2**	13.5	RLM1916	
	19	20.2	0.5	**FIR-151920-2**	16.8	RLM1920	
	20	10.2	0.5	**FIR-152010-2**	10.9	RLM2010	
	20	15.2	0.5	**FIR-152015-2**	16.3	RLM2015	
	20	20.2	0.5	**FIR-152020-2**	21.7	RLM2020	RLM202820
	20	25.2	0.5	**FIR-152025-2**	27	RLM2025	
17	21	16.2	0.5	**FIR-172116-2**	15.1	RLM2116	
	21	20.2	0.5	**FIR-172120-2**	18.8	RLM2120	
	22	25.2	0.5	**FIR-172215-2**	18.2	RLM2215	
	22	16.2	0.5	**FIR-172216-2**	19.4		RLM223016
	22	20.2	0.5	**FIR-172220-2**	24.1	RLM2220	RLM223020
	22	25.2	0.5	**FIR-172225-2**	30.1	RLM2225	
20	24	16.2	0.5	**FIR-202416-2**	17.5		RLM243216
	24	20.2	0.5	**FIR-202420-2**	21.8	RLM2420	
	24	28.2	0.5	**FIR-202428-2**	30.4	RLM2428	
	25	12.2	0.5	**FIR-202512-2**	16.8	RLM2512	
	25	20.2	0.5	**FIR-202520-2**	27.8	RLM2520	RLM253320
	25	25.2	0.5	**FIR-202525-2**	34.7	RLM2525	RLM253325

FIR

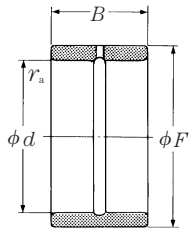
Bore Diameter 22 — 50 mm



Boundary Dimensions (mm)				Inner Ring Numbers	Mass (g)	Matching Solid Needle Roller Bearings	
						RLM	
d	F	B	r min		approx		
22	26	20.2	0.5	**FIR-222620-2**	23.8	RLM2620	
	28	20.2	0.5	**FIR-222820-2**	37.1	RLM2820	RLM283720
	28	25.2	0.5	**FIR-222825-2**	46.3	RLM2825	
	28	30.2	0.5	**FIR-222830-2**	55.5		RLM283730
25	29	20.2	0.5	**FIR-252920-2**	26.7	RLM2920	
	29	30.2	0.5	**FIR-252930-2**	40	RLM2930	
	30	15.2	0.5	**FIR-253015-2**	25.6	RLM3015	
	30	20.2	0.5	**FIR-253020-2**	34	RLM3020	RLM304020
	30	25.2	0.5	**FIR-253025-2**	42.5	RLM3025	RLM304025
28	32	20.2	0.5	**FIR-283220-2**	29.7	RLM3220	
	32	30.2	0.5	**FIR-283230-2**	44.4	RLM3230	
30	35	20.2	0.5	**FIR-303520-2**	40.2	RLM3520	RLM354520
	35	25.2	0.5	**FIR-303525-2**	50.2		RLM354525
	35	30.2	0.5	**FIR-303530-2**	60.1	RLM3530	RLM354530
32	37	20.3	0.6	**FIR-323720-2**	42.9	RLM3720	
	37	30.3	0.6	**FIR-323730-2**	64	RLM3730	
	38	15.3	0.6	**FIR-323815-2**	39.4	RLM3815	
	38	20.3	0.6	**FIR-323820-2**	52.2	RLM3820	
	38	25.3	0.6	**FIR-323825-2**	65.1	RLM3825	
	38	30.3	0.6	**FIR-323830-2**	78	RLM3830	
35	40	15.3	0.6	**FIR-354015-2**	35.1	RLM4015	
	40	20.3	0.6	**FIR-354020-2**	46.6	RLM4020	
	40	25.3	0.6	**FIR-354025-2**	58.1	RLM4025	
	40	30.3	0.6	**FIR-354030-2**	69.6	RLM4030	
40	45	20.3	0.6	**FIR-404520-2**	52.9	RLM4520	
	45	25.3	0.6	**FIR-404525-2**	65.9	RLM4525	
	45	30.3	0.6	**FIR-404530-2**	78.9	RLM4530	
42	50	20.3	0.6	**FIR-425020-2**	91.5	RLM5020	RLM506220
	50	25.3	0.6	**FIR-425025-2**	114		RLM506225
	50	30.3	0.6	**FIR-425030-2**	137	RLM5030	RLM506230
45	55	30.3	0.6	**FIR-455530-2**	186	RLM5530	
	55	40.3	0.6	**FIR-455540-2**	247	RLM5540	
50	60	30.3	0.6	**FIR-506030-2**	204		RLM607230
	60	40.3	0.6	**FIR-506040-2**	272	RLM6040	RLM607240

IR

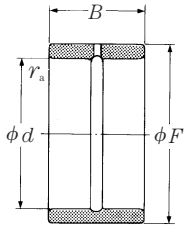
Bore Diameter 9.525 — 39.688 mm



Boundary Dimensions (mm, inch) d		F		B	Inner Ring Numbers	Mass (kg) approx	Fillet Radius of Shaft (mm) r_a max	Matching Solid Needle Roller Bearings HJ
9.525	0.3750	15.875	0.6250	19.05	**IR-061012**	0.018	0.6	HJ-101812
12.700	0.5000	19.050	0.7500	19.05	**IR-081212**	0.027	1	HJ-122012
		19.050	0.7500	25.40	**IR-081216**	0.037	1	HJ-122016
15.875	0.6250	22.225	0.8750	19.05	**IR-101412**	0.026	1	HJ-142212
		22.225	0.8750	25.40	**IR-101416**	0.033	1	HJ-142216
17.462	0.6875	22.225	0.8750	19.05	**IR-111412**	0.023	1	HJ-142212
19.050	0.7500	25.400	1.0000	19.05	**IR-121612**	0.032	1	HJ-162412
		25.400	1.0000	25.40	**IR-121616**	0.044	1	HJ-162416
20.638	0.8125	25.400	1.0000	25.40	**IR-131616**	0.034	1	HJ-162416
22.225	0.8750	28.575	1.1250	25.40	**IR-141816**	0.050	1	HJ-182616
		28.575	1.1250	31.75	**IR-141820**	0.065	1	HJ-182620
23.812	0.9375	28.575	1.1250	25.40	**IR-151816**	0.040	1	HJ-182616
		28.575	1.1250	31.75	**IR-151820**	0.050	1	HJ-182620
25.400	1.0000	31.750	1.2500	25.40	**IR-162016**	0.060	1	HJ-202816
		31.750	1.2500	31.75	**IR-162020**	0.075	1	HJ-202820
28.575	1.1250	34.925	1.3750	25.40	**IR-182216**	0.065	1	HJ-223016
		34.925	1.3750	31.75	**IR-182220**	0.075	1	HJ-223020
30.162	1.1875	38.100	1.5000	31.75	**IR-192420**	0.105	1.5	HJ-243320
31.750	1.2500	38.100	1.5000	25.40	**IR-202416**	0.075	1.5	HJ-243316
		38.100	1.5000	31.75	**IR-202420**	0.090	1.5	HJ-243320
33.338	1.3125	41.275	1.6250	25.40	**IR-212616**	0.095	1.5	HJ-263516
		41.275	1.6250	31.75	**IR-212620**	0.12	1.5	HJ-263520
34.925	1.3750	41.275	1.6250	31.75	**IR-222620**	0.10	1.5	HJ-263520
		44.450	1.7500	31.75	**IR-222820**	0.145	1.5	HJ-283720
36.512	1.4375	44.450	1.7500	25.40	**IR-232816**	0.10	1.5	HJ-283716
		44.450	1.7500	31.75	**IR-232820**	0.125	1.5	HJ-283720
38.100	1.5000	44.450	1.7500	25.40	**IR-242816**	0.095	1.5	HJ-283716
		44.450	1.7500	31.75	**IR-242820**	0.10	1.5	HJ-283720
		50.800	2.0000	31.75	**IR-243220**	0.22	1.5	HJ-324120
39.688	1.5625	47.625	1.8750	31.75	**IR-253020**	0.135	1.5	HJ-303920
		50.800	2.0000	31.75	**IR-253220**	0.195	1.5	HJ-324120

IR

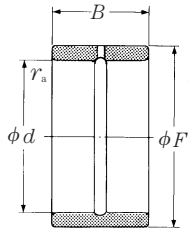
Bore Diameter 41.275 — 107.950 mm



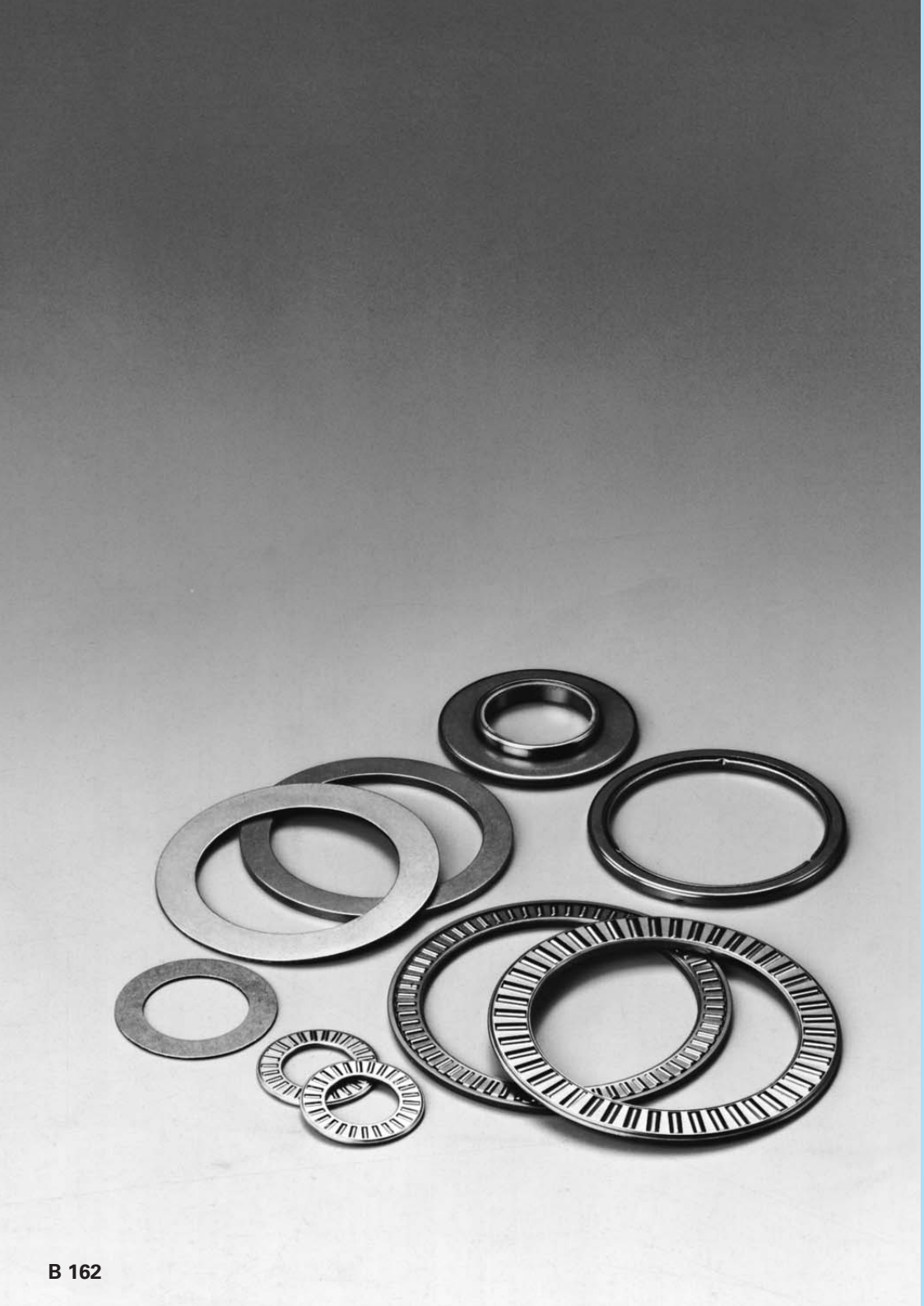
Boundary Dimensions (mm, inch) d		F		B	Inner Ring Numbers	Mass (kg) approx	Fillet Radius of Shaft (mm) r_a max	Matching Solid Needle Roller Bearings HJ
41.275	1.6250	50.800	2.0000	31.75	**IR-263220**	0.17	1.5	HJ-324120
42.862	1.6875	50.800	2.0000	25.40	**IR-273216**	0.12	1.5	HJ-324116
		50.800	2.0000	31.75	**IR-273220**	0.145	1.5	HJ-324120
44.450	1.7500	57.150	2.2500	38.10	**IR-283624**	0.305	1.5	HJ-364824
		57.150	2.2500	44.45	**IR-283628**	0.355	1.5	HJ-364828
49.212	1.9375	63.500	2.5000	38.10	**IR-314024**	0.375	2	HJ-405224
		63.500	2.5000	44.45	**IR-314028**	0.425	2	HJ-405228
50.800	2.0000	63.500	2.5000	38.10	**IR-324024**	0.34	2	HJ-405224
		63.500	2.5000	44.45	**IR-324028**	0.40	2	HJ-405228
55.562	2.1875	69.850	2.7500	44.45	**IR-354428**	0.50	2	HJ-445628
57.150	2.2500	69.850	2.7500	38.10	**IR-364424**	0.375	2	HJ-445624
		69.850	2.7500	45.45	**IR-364428**	0.45	2	HJ-445628
60.325	2.3750	76.200	3.0000	44.45	**IR-384828**	0.57	2	HJ-486028
63.500	2.5000	76.200	3.0000	38.10	**IR-404824**	0.44	2	HJ-486024
		76.200	3.0000	44.45	**IR-404828**	0.47	2	HJ-486028
69.850	2.7500	82.550	3.2500	44.45	**IR-445228**	0.50	2	HJ-526828
		82.550	3.2500	50.80	**IR-445232**	0.60	2	HJ-526832
74.612	2.9375	88.900	3.5000	50.80	**IR-475632**	0.75	2	HJ-567232
76.200	3.0000	88.900	3.5000	50.80	**IR-485632**	0.70	2	HJ-567232
79.375	3.1250	95.250	3.7500	50.80	**IR-506032**	0.90	2.5	HJ-607632
82.550	3.2500	95.250	3.7500	50.80	**IR-526032**	0.70	2.5	HJ-607632
		101.600	4.0000	50.80	**IR-526432**	1.1	2.5	HJ-648032
85.725	3.3750	101.600	4.0000	50.80	**IR-546432**	0.90	2.5	HJ-648032
88.900	3.5000	101.600	4.0000	50.80	**IR-566432**	0.75	2.5	HJ-648032
		107.950	4.2500	50.80	**IR-566832**	1.15	2.5	HJ-688432
95.250	3.7500	107.950	4.2500	50.80	**IR-606832**	1.0	2.5	HJ-688432
		114.300	4.5000	57.15	**IR-607236**	1.4	2.5	HJ-729636
		114.300	4.5000	63.50	**IR-607240**	1.55	2.5	HJ-729640
101.600	4.0000	127.000	5.0000	57.15	**IR-648036**	2.05	2.5	HJ-8010436
		127.000	5.0000	63.50	**IR-648040**	2.25	2.5	HJ-8010440
107.950	4.2500	127.000	5.0000	57.15	**IR-688036**	1.55	2.5	HJ-8010436

IR

Bore Diameter 114.300 — 203.200 mm



Boundary Dimensions (mm, inch) d		F		B	Inner Ring Numbers	Mass (kg) approx	Fillet Radius of Shaft (mm) r_a max	Matching Solid Needle Roller Bearings HJ
114.300	4.5000	139.700	5.5000	63.50	**IR-728840**	2.5	2.5	HJ-8811240
		139.700	5.5000	76.20	**IR-728848**	2.95	2.5	HJ-8811248
120.650	4.7500	146.050	5.7500	76.20	**IR-769248**	3.15	3	HJ-9211648
127.000	5.0000	152.400	6.0000	63.50	**IR-809640**	2.75	3	HJ-9612040
		152.400	6.0000	76.20	**IR-809648**	3.35	3	HJ-9612048
139.700	5.5000	165.100	6.5000	63.50	**IR-8810440**	3.05	3	HJ-10412840
		165.100	6.5000	76.20	**IR-8810448**	3.65	3	HJ-10412848
152.400	6.0000	184.150	7.2500	76.20	**IR-9611648**	4.9	3	HJ-11614648
165.100	6.5000	196.850	7.7500	76.20	**IR-10412448**	5.5	3	HJ-12415448
177.800	7.0000	209.550	8.2500	76.20	**IR-11213248**	5.55	3	HJ-13216248
190.500	7.5000	222.250	8.7500	76.20	**IR-12014048**	6.15	4	HJ-14017048
203.200	8.0000	234.950	9.2500	76.20	**IR-12814848**	6.4	4	HJ-14817848



THRUST BEARINGS

CAGE & NEEDLE ROLLER ASSEMBLIES FOR THRUST NEEDLE BEARINGS

			Pages
Metric Design	Bore Diameter	10–100 mm	B168–B169
Inch Design	Bore Diameter	6.350–104.775 mm	B170–B171

THRUST NEEDLE BEARINGS WITH INTEGRATED RACEWAY WASHERS

Metric Design (FB)	Bore Diameter	20–55 mm	B172–B173
Metric Design (FH, FG)	Bore Diameter	22.6–43 mm	B174–B175

Thrust needle bearings type comes in metric and inch designs.

The cage & Needle roller assembly for the thrust needle bearing is a cage that has two pieces joined together (refer to **Fig. 1**), to which surface hardening is applied after the precision press processing of a steel plate, and in which high accuracy rollers are incorporated, in order to achieve rigidity and wear resistance. Many rollers are retained securely in the cage, taking large axial load, and yet rotating smoothly. Mounting space is so small that it is easy to replace a conventional sliding thrust washer. Different thicknesses of various types of thrust raceway washers are available in this cage & Needle roller assembly, which allows selection of the appropriate thickness suited for bearing peripheral mounting conditions.

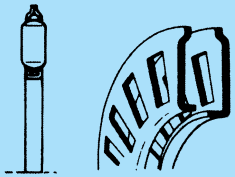


Fig. 1

Type

Several types of thrust bearings are shown in **Table 1**.

For inch bearings, the dimension number is expressed in integers in units of 1/16 inch.

Raceway washers are available for use when the raceway surface cannot be hardened by heat treatment and ground to the proper finish.

Types of raceway washers used for cage & Needle roller assemblies in thrust needle bearings are shown in **Table 2**.

Table 1 Bearing Types

Code		Description
Metric	FNTA	Cage & Needle roller assemblies for thrust needle bearings
	FB, FG, FH	Thrust needle bearings with integrated raceway washers
Inch	NTA, NTC	Cage & Needle roller assemblies for thrust needle bearings

The composition of bearing numbers is as follows:

(Example) FNTA-2035

- Outside Diameter (mm)
- Bore Diameter (mm)
- Type Code

Table 2 Raceway Washers of Thrust Needle Bearings

Units: mm

Metric		Inch	
Type	Thickness	Type	Thickness
FTRA	1.0	TRA	0.8
FTRB	1.5	TRB	1.6
FTRC	2.0	TRC	2.4
FTRD	2.5	TRD	3.2
FTRE	3.0	TRE	4.0
—	—	TRF	4.8

Other than the rings shown in **Table 2**, raceway washers with flanges at the bore or outside are also available.

Accuracy

Thrust Needle Bearings

NSK metric FNTA cage & Needle roller assemblies conform with ISO criteria, and the fitting tolerance of the cage born diameter is E11 and that of the outside diameter is c12.

The dimensional tolerances for inch NTA cage & Needle roller assemblies, and metric and inch thrust raceway washers are shown in the dimensional table. The accuracy class of the rollers is Class 3 of ISO.

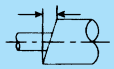
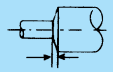
Application

Thrust Needle Bearings

When using FNTA and NTA cage & Needle roller assemblies, using the gear side as a raceway ring saves bearing space (refer to **Fig. 2**).

In general, the accuracy, surface roughness, and hardness of the raceway components should satisfy the specifications shown in **Table 3**.

Table 3 Specifications of Thrust Bearing Raceway Surface

Squareness A	0.5/1000 (mm/mm)	
Squareness B	1.0/1000 (mm/mm)	
Roughness, (1) Ra	0.4	—
Hardness, (2)	HRC 58 to 64 (HRC60 to 64 is favorable)	—

Notes (1) Refer to Section **5.2** on page A34 for details.
(2) Refer to Section **5.3** on page A35 for details.

When a cage & Needle roller assembly cannot be used by itself, a thrust raceway washer should be employed. The raceway washer should be selected after considering washer support face and the volume of the load. The thrust raceway washer should be set to prevent sliding between the contact surfaces.

When the area supporting the raceway washer is sufficiently large, a thin raceway washer, such as the FTRA and TRA types, can be used. It should be noted that even if the washers of the FTRA and TRA types are warped, an axial load of more than approximately 196 N {20 kgf} will flatten them.

When the support face of the shaft or housing is small (refer to **Fig. 3**). FTRE and TRD types, which are thick-wall raceway washers, or any of the other thicker types should be used. In any case, to bring out the full performance of the bearing, the squareness of the support face should be ensured and sufficient space is needed in the support face.

For squareness of the raceway washer support face, the values shown in **Table 3** are applied. Please consult NSK regarding shape and accuracy.

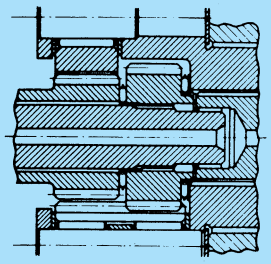


Fig. 2

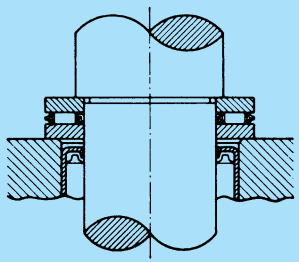


Fig. 3

Bearing Guide

The cages used in FNTA and NTA cage & Needle roller assemblies are guided by the shaft along their extended cage bore.

Bore diameter tolerance is smaller than that for the outside diameter, and as relative speed at the guide face is slow, the bore guiding method is effective. When the outside of the cage is guided, it is necessary for the housing guide surface to be hardened, at least on its surface, to protect it from wear.

Fitting

Recommended fittings for thrust needle bearings is shown in **Table 4**.

Table 4 Recommended Fitting for Thrust Bearing

Units: mm

Classification	Type	Cage or Raceway Washer Guide	Fitting Class or Tolerance	
			Shaft	Housing Bore
Thrust Needle Bearing Cage & Needle Roller Assembly	FNTA	Bore	h8	D_c+over 1.0
		Outside	—	H10
	NTA NTC	Bore	$^{0}_{-0.075}$	D_c+over 1.0
		Outside	—	H10
Thrust Needle Bearing Raceway Washer	FTRA — FTRE	Bore	h8	D_c+over 1.0
		Outside	—	H10
	TRA — TRF	Bore	$^{0}_{-0.075}$	D_c+over 1.0
		Outside	—	H10

Remarks D_c in the table to the bore and outside surface of the cage.
Please consult NSK regarding fitting of bearing types other than those shown in **Table 4**.

Thrust Needle Bearings

FNTA (Thrust Cage & Needle Roller Assemblies)

FTRA (Thrust Bearing Rings s = 1.0)

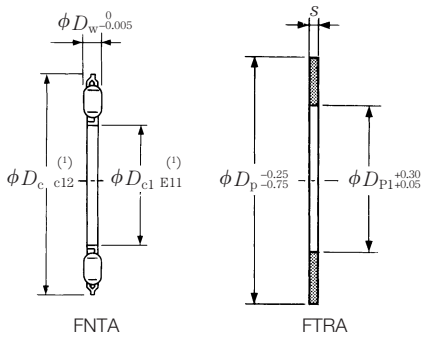
FTRB (s = 1.5)

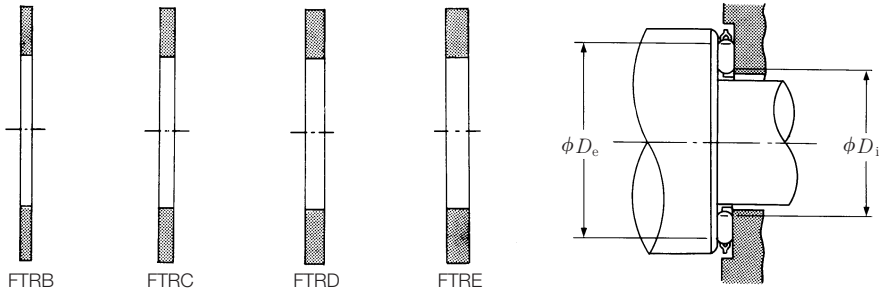
FTRC (s = 2.0)

FTRD (s = 2.5)

FTRE (s = 3.0)

Bore Diameter 10 — 100 mm





Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers
D_{C1},D_{P1}	D_{C},D_{P}	D_{W}	C_{a}	C_{oa}	C_{a}	C_{oa}	Oil	
10	24	2	7 750	23 000	790	2 350	17 000	**FNTA-1024**
12	26	2	8 350	26 300	855	2 680	16 000	**FNTA-1226**
13	26	2	8 050	25 400	825	2 590	16 000	**FNTA-1326**
15	28	2	7 950	25 800	810	2 630	15 000	**FNTA-1528**
16	29	2	8 200	27 100	835	2 770	14 000	**FNTA-1629**
17	30	2	8 400	28 400	855	2 900	14 000	**FNTA-1730**
18	31	2	8 600	29 700	875	3 050	13 000	**FNTA-1831**
20	35	2	11 900	47 000	1 220	4 800	12 000	**FNTA-2035**
25	42	2	14 800	66 000	1 510	6 750	9 500	**FNTA-2542**
30	47	2	16 500	79 000	1 680	8 100	8 500	**FNTA-3047**
35	52	2	17 300	88 000	1 770	8 950	8 000	**FNTA-3552**
40	60	3	26 900	122 000	2 740	12 400	6 700	**FNTA-4060**
45	65	3	28 700	137 000	2 930	14 000	6 300	**FNTA-4565**
50	70	3	30 500	152 000	3 100	15 500	5 600	**FNTA-5070**
55	78	3	37 000	201 000	3 750	20 500	5 300	**FNTA-5578**
60	85	3	43 000	252 000	4 400	25 700	4 800	**FNTA-6085**
65	90	3	45 500	274 000	4 600	28 000	4 500	**FNTA-6590**
70	95	4	59 000	320 000	6 000	33 000	4 300	**FNTA-7095**
75	100	4	60 000	335 000	6 150	34 500	4 000	**FNTA-75100**
80	105	4	63 000	365 000	6 450	37 500	3 800	**FNTA-80105**
85	110	4	64 500	380 000	6 550	39 000	3 600	**FNTA-85110**
90	120	4	80 000	515 000	8 150	52 500	3 400	**FNTA-90120**
100	135	4	98 500	695 000	10 000	71 000	3 000	**FNTA-100135**

Matching Bearing Rings					Roller Contact Surfaces (mm)		Mass approx (g)	
$s=1.0^{\pm 0.05}$	$s=1.5^{0}_{-0.08}$	$s=2.0^{0}_{-0.08}$	$s=2.5^{0}_{-0.08}$	$s=3.0^{0}_{-0.08}$	D_{e} min	D_{i} max	FNTA	FTRA
*FTRA-1024	FTRB-1024	FTRC-1024	—	—	22	11.5	2.3	2.9
FTRA-1226	FTRB-1226	FTRC-1226	—	—	24	13.5	3.4	3.3
FTRA-1326	FTRB-1326	FTRC-1326	—	—	24	14.5	3.3	3.1
FTRA-1528	FTRB-1528	FTRC-1528	FTRD-1528	FTRE-1528	26	16.5	3.5	3.5
FTRA-1629	FTRB-1629	FTRC-1629	FTRD-1629	FTRE-1629	27	17.5	3.7	3.6
FTRA-1730	FTRB-1730	FTRC-1730	FTRD-1730	FTRE-1730	28	18.5	3.8	3.8
FTRA-1831	FTRB-1831	FTRC-1831	FTRD-1831	FTRE-1831	29	19.5	4.0	3.9
FTRA-2035	FTRB-2035	FTRC-2035	FTRD-2035	FTRE-2035	33	21.5	5.4	5.1
FTRA-2542	FTRB-2542	FTRC-2542	FTRD-2542	FTRE-2542	40	26.5	7.7	7.0
FTRA-3047	FTRB-3047	FTRC-3047	FTRD-3047	FTRE-3047	45	31.5	8.9	7.9
FTRA-3552	FTRB-3552	FTRC-3552	FTRD-3552	FTRE-3552	50.5	36.5	9.7	9.1
FTRA-4060	FTRB-4060	FTRC-4060	FTRD-4060	FTRE-4060	57	42	18	12
FTRA-4565	FTRB-4565	FTRC-4565	FTRD-4565	FTRE-4565	62	47	20	13
FTRA-5070	FTRB-5070	FTRC-5070	FTRD-5070	FTRE-5070	67	51.5	22	15
FTRA-5578	FTRB-5578	FTRC-5578	FTRD-5578	FTRE-5578	75	57	29	19
FTRA-6085	FTRB-6085	FTRC-6085	FTRD-6085	FTRE-6085	82	61.5	35	22
FTRA-6590	FTRB-6590	FTRC-6590	FTRD-6590	FTRE-6590	87.5	66.5	38	24
FTRA-7095	FTRB-7095	FTRC-7095	FTRD-7095	FTRE-7095	92.5	71.5	52	25
FTRA-75100	FTRB-75100	FTRC-75100	FTRD-75100	FTRE-75100	97.5	76.5	54	27
FTRA-80105	FTRB-80105	FTRC-80105	FTRD-80105	FTRE-80105	102.5	81.5	58	28
FTRA-85110	FTRB-85110	FTRC-85110	FTRD-85110	FTRE-85110	107.5	86.5	63	30
FTRA-90120	FTRB-90120	FTRC-90120	FTRD-90120	FTRE-90120	117.5	91.5	80	38
FTRA-100135	FTRB-100135	FTRC-100135	FTRD-100135	FTRE-100135	132.5	101.5	105	50

Note (1) For tolerance classes c12 and E11, please refer to pages C58 and C60, respectively.

* The tolerance of this bearing's bore diameter is +0.025 to +0.175 mm and its outside diameter tolerance is −0.040 to −0.370 mm.

Thrust Needle Bearings

NTA (Thrust Cage & Needle Roller Assemblies)

NTC

TRA (Thrust Bearing Rings s = 0.8)

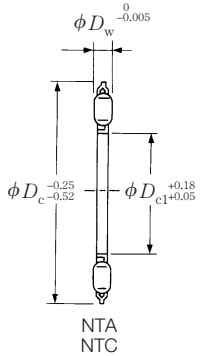
TRB (s = 1.6)

TRC (s = 2.4)

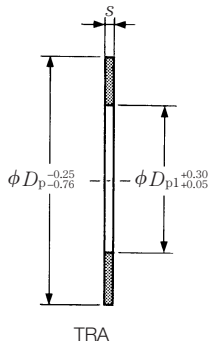
TRD (s = 3.2)

TRE (s = 4.0)

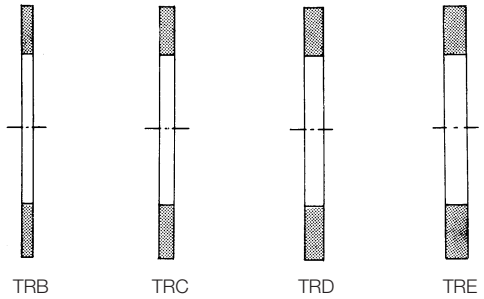
Bore Diameter 6.350 — 104.775 mm



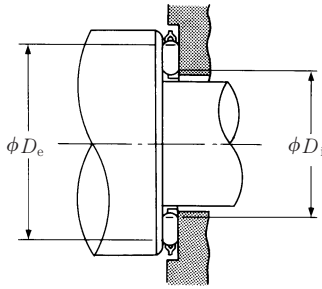
NTA
NTC



TRA



TRB TRC TRD TRE



Boundary Dimensions (mm, inch)						Basic Load Ratings (N)		{kgf}		Limiting Speeds (min^{-1})
D_{C1}, D_{P1}		D_C, D_P		D_W		C_a	C_{oa}	C_a	C_{oa}	Oil
6.350	0.2500	17.450	—	1.984	—	4 400	9 900	450	1 010	24 000
7.925	—	19.050	0.7500	1.984	—	4 950	12 100	505	1 230	22 000
9.525	0.3750	20.625	—	1.984	—	5 200	13 200	530	1 350	20 000
12.700	0.5000	23.800	—	1.984	—	6 150	17 600	625	1 790	18 000
14.275	—	25.400	1.0000	1.984	—	6 600	19 800	670	2 020	17 000
15.875	0.6250	28.575	1.1250	1.984	—	8 100	26 600	825	2 720	15 000
19.050	0.7500	31.750	1.2500	1.984	—	9 000	32 000	915	3 250	13 000
22.225	0.8750	36.500	—	1.984	—	11 600	46 500	1 180	4 750	11 000
		42.850	—	1.984	—	16 300	74 500	1 660	7 600	9 500
25.400	1.0000	39.675	—	1.984	—	12 000	50 000	1 220	5 100	10 000
28.575	1.1250	44.450	1.7500	1.984	—	14 800	68 000	1 510	6 950	9 000
31.750	1.2500	49.200	—	1.984	—	17 700	88 500	1 800	9 050	8 000
34.925	1.3750	52.375	—	1.984	—	18 700	98 000	1 910	10 000	7 500
38.100	1.5000	55.550	—	1.984	—	20 400	112 000	2 080	11 400	7 500
44.450	1.7500	63.500	2.5000	1.984	—	20 400	117 000	2 080	11 900	6 700
50.800	2.0000	69.850	2.7500	1.984	—	21 100	126 000	2 150	12 800	6 000
53.975	2.1250	73.025	2.8750	1.984	—	21 400	131 000	2 190	13 300	5 600
57.150	2.2500	76.200	3.0000	1.984	—	21 800	135 000	2 220	13 800	5 300
		79.375	3.1250	3.175	0.1250	33 500	172 000	3 450	17 500	5 000
63.500	2.5000	82.550	3.2500	1.984	—	22 400	145 000	2 290	14 700	5 000
69.850	2.7500	92.075	3.6250	3.175	0.1250	42 500	248 000	4 350	25 300	4 500
76.200	3.0000	95.250	3.7500	1.984	—	23 700	163 000	2 410	16 600	4 300
82.550	3.2500	104.775	4.1250	3.175	0.1250	46 000	287 000	4 700	29 200	3 800
95.250	3.7500	117.475	4.6250	3.175	0.1250	50 000	335 000	5 100	34 000	3 400
104.775	4.1250	128.575	—	3.175	0.1250	57 000	405 000	5 800	41 000	3 200

Bearing Numbers	Matching Bearing Rings					Roller Contact Surfaces (mm)		Mass (g)	
	$s=0.8\,^{+0.01}_{-0.04}$	$s=1.6\,^{0}_{-0.08}$	$s=2.4\,^{+0.01}_{-0.07}$	$s=3.2\,^{0}_{-0.08}$	$s=4.0\,^{-0.01}_{-0.09}$	D_e min	D_i max	approx NTA	TRA
NTA-411	**TRA-411**	**TRB-411**	**TRC-411**	—	—	16	7.5	1.4	1.4
NTA-512	**TRA-512**	**TRB-512**	—	—	—	17.5	9	1.8	1.4
NTA-613	**TRA-613**	**TRB-613**	**TRC-613**	—	—	19	10.5	1.8	1.4
NTA-815	**TRA-815**	**TRB-815**	**TRC-815**	—	—	22	14	2.3	1.8
NTA-916	**TRA-916**	**TRB-916**	**TRC-916**	—	—	23.5	15.5	2.7	2.3
NTA-1018	**TRA-1018**	**TRB-1018**	**TRC-1018**	**TRD-1018**	**TRE-1018**	27	17	3.2	2.7
NTA-1220	**TRA-1220**	**TRB-1220**	**TRC-1220**	**TRD-1220**	**TRE-1220**	30	20	4.1	3.2
NTA-1423	**TRA-1423**	**TRB-1423**	**TRC-1423**	**TRD-1423**	—	34.5	23.5	5.0	4.1
NTC-1427	—	**TRB-1427**	**TRC-1427**	**TRD-1427**	—	41	25	9.1	—
NTA-1625	**TRA-1625**	**TRB-1625**	**TRC-1625**	**TRD-1625**	**TRE-1625**	38	26.5	5.9	4.5
NTA-1828	**TRA-1828**	**TRB-1828**	**TRC-1828**	**TRD-1828**	—	42.5	29.5	8.6	5.9
NTA-2031	**TRA-2031**	**TRB-2031**	**TRC-2031**	**TRD-2031**	—	47.5	33	9.5	6.8
NTA-2233	**TRA-2233**	**TRB-2233**	**TRC-2233**	**TRD-2233**	**TRE-2233**	50.5	36	10	7.3
NTA-2435	**TRA-2435**	**TRB-2435**	**TRC-2435**	**TRD-2435**	—	53.5	39	11	7.7
NTA-2840	**TRA-2840**	**TRB-2840**	**TRC-2840**	**TRD-2840**	—	60	45.5	14	9.5
NTA-3244	**TRA-3244**	**TRB-3244**	**TRC-3244**	**TRD-3244**	—	66.5	52	15	11
NTA-3446	**TRA-3446**	**TRB-3446**	**TRC-3446**	**TRD-3446**	—	69.5	55	16	12
NTA-3648	**TRA-3648**	**TRB-3648**	**TRC-3648**	**TRD-3648**	—	72.5	58.5	17	12
NTA-3650 (1)	—	—	—	—	—	76.5	58.5	29	—
NTA-4052	**TRA-4052** (2)	**TRB-4052** (2)	**TRC-4052** (2)	**TRD-4052** (2)	—	79	64.5	19	14
NTA-4458 (1)	**TRA-4458** (2)	**TRB-4458** (2)	**TRC-4458** (2)	**TRD-4458** (2)	—	89	71.5	37	18
NTA-4860	**TRA-4860** (2)	**TRB-4860** (2)	—	**TRD-4860** (2)	—	92	77.5	22	16
NTA-5266 (1)	**TRA-5266** (2)	—	—	**TRD-5266** (2)	—	102	84	42	20
NTA-6074 (1)	**TRA-6074** (2)	**TRB-6074** (2)	**TRC-6074** (2)	**TRD-6074** (2)	—	114.5	97	50	23
NTA-6681 (1)	**TRA-6681** (2)	—	**TRC-6681** (2)	**TRD-6681** (2)	—	125.5	106.5	62	27

Notes (1) Tolerance of cage bore diameter is +0.05 to +0.25mm, and tolerance of cage outside diameter is −0.25 to −0.64mm.

(2) Tolerance of bearing ring bore diameter is +0.05 to −0.43mm.

Thrust Needle Bearings

FB (With Integrated Raceway Washers)

FTRA (Thrust Bearing Rings s = 1.0)

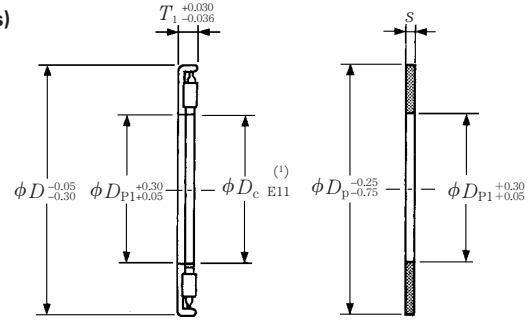
FTRB (s = 1.5)

FTRC (s = 2.0)

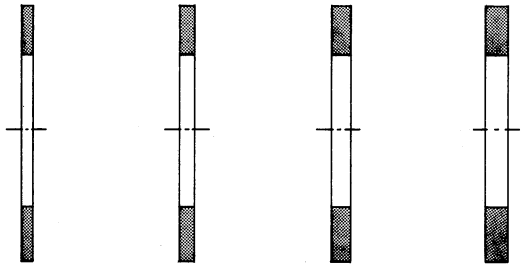
FTRD (s = 2.5)

FTRE (s = 3.0)

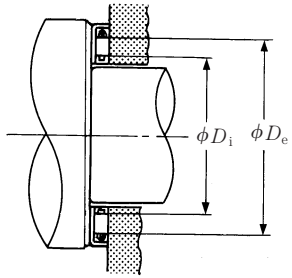
Bore Diameter 20 — 55 mm



FB FTRA



FTRB FTRC FTRD FTRE



Boundary Dimensions (mm)			Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers
D_C,D_{P1}	D_C,D_P	T_1	C_a	C_{oa}	C_a	C_{oa}	Oil	
20	35	2.8	9 450	34 500	965	3 550	13 000	**FB502002**
30	47	2.8	12 700	56 500	1 300	5 750	9 000	**FB503002**
35	53	2.8	15 600	77 000	1 590	7 850	8 000	**FB503503**
40	60	2.8	20 200	112 000	2 060	11 400	7 100	**FB504001**
45	65	2.8	20 400	117 000	2 080	11 900	6 300	**FB504501**
50	70	2.8	21 200	126 000	2 160	12 800	6 000	**FB505002**
55	75	2.8	22 500	140 000	2 290	14 300	5 600	**FB505501**

Matching Bearing Rings					Roller Contact Surfaces (mm)		Mass (g)
$s=1.0^{\pm 0.05}$	$s=1.5^{\ 0}_{-0.08}$	$s=2.0^{\ 0}_{-0.08}$	$s=2.5^{\ 0}_{-0.08}$	$s=3.0^{\ 0}_{-0.08}$	D_e min	D_i max	approx
FTRA-2035	**FTRB-2035**	**FTRC-2035**	**FTRD-2035**	**FTRE-2035**	31	21	13
FTRA-3047	**FTRB-3047**	**FTRC-3047**	**FTRD-3047**	**FTRE-3047**	43	31	16
FTRA-3553	**FTRB-3553**	**FTRC-3553**	**FTRD-3553**	**FTRE-3553**	48.5	36.5	19
FTRA-4060	**FTRB-4060**	**FTRC-4060**	**FTRD-4060**	**FTRE-4060**	55	41	25
FTRA-4565	**FTRB-4565**	**FTRC-4565**	**FTRD-4565**	**FTRE-4565**	60.5	46	27
FTRA-5070	**FTRB-5070**	**FTRC-5070**	**FTRD-5070**	**FTRE-5070**	65.5	51	30
FTRA-5575	**FTRB-5575**	**FTRC-5575**	**FTRD-5575**	**FTRE-5575**	70.5	56	32

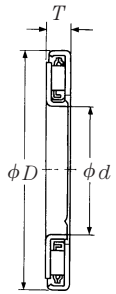
Note (1) For tolerance classes E11, please refer to pages C60.

Thrust Needle Bearings

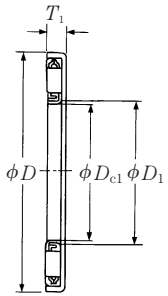
FH (With Integrated Raceway Washers)

FG

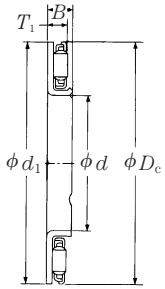
Bore Diameter 22.6 — 43 mm



Example 1



Example 2



Example 3

Boundary Dimensions (mm)						Basic Load Ratings (N)		Basic Load Ratings {kgf}		Limiting Speeds (min^{-1})	Bearing Numbers	Example	Mass (g)
d,D_1	D_{c1}	D,d_1	D_c	T,T_1	B	C_a	C_{oa}	C_a	C_{oa}	Oil			approx
22.6	—	38.7	—	3.894	—	10 100	40 000	1 030	4 100	11 000	**FH502201A**	1	16.5
27.7	—	45.9	—	3.604	—	15 300	71 000	1 560	7 250	9 000	**FH502702A**	1	20.2
36	—	53	—	3.8	—	13 700	66 000	1 400	6 700	8 000	**FH503604-1**	1	24.1
39.35	—	54.02	—	3.604	—	11 400	52 500	1 160	5 350	7 500	**FH503901A**	1	20.5
48.8	—	70	—	4	—	20 300	120 000	2 070	12 200	6 000	**FH504801**	1	42.4
58.5	—	78	—	4	—	20 400	125 000	2 080	12 800	5 300	**FH505802**	1	44.6
64.4	—	78.1	—	4	—	10 200	52 500	1 040	5 350	5 300	**FH506401**	1	32.7
26.2	28.575	46.32	—	3.81	—	19 800	74 500	2 020	7 600	9 000	**FG502602**	2	20.2
30	30	47	—	2.824	—	12 700	56 500	1 300	5 750	9 000	**FG503006**	2	14.8
35	35	53	—	2.824	—	15 600	77 000	1 590	7 850	8 000	**FG503502**	2	18.2
38.5	38	57.3	—	2.81	—	15 600	78 000	1 590	7 950	7 500	**FG503804**	2	20.9
42	42	61.98	—	3.81	—	24 700	110 000	2 520	11 200	6 700	**FG504201**	2	31.7
45	45	64	—	2.8	—	20 400	117 000	2 080	11 900	6 300	**FG504501**	2	23.2
48.4	48.4	68.48	—	3.81	—	25 000	117 000	2 550	11 900	6 000	**FG504801**	2	35.5
52	52	72.1	—	2.81	—	21 000	126 000	2 140	12 800	5 600	**FG505203**	2	28.1
58	58	78	—	2.8	—	23 300	149 000	2 380	15 200	5 300	**FG505801**	2	31.8
19	—	45	45	2.794	3.43	13 200	60 000	1 340	6 100	9 000	**FG501903**	3	19.8
28	—	52.5	52	2.81	3.43	17 200	88 000	1 760	8 950	7 500	**FG502801**	3	23
30.5	—	48	48	2.794	3.43	13 400	63 000	1 370	6 450	8 500	**FG503002**	3	16.3
33.5	—	47.8	48	2.794	3.43	13 400	63 000	1 370	6 450	8 500	**FG503301**	3	14.1
43	—	60	60.05	2.8	3.43	16 900	91 500	1 720	9 350	6 700	**FG504302**	3	20.9



CAM FOLLOWERS (STUD TYPE TRACK ROLLERS)

			Pages
Metric Design	Outside Diameter	10－90 mm	B182－B185
Inch Design	Outside Diameter	12.700－152.400 mm	B186－B189

Cam followers, which are equipped with a thick-wall outer ring, crowned rollers, and a stud with a hardened raceway surface, are especially tough and can sustain shock loads.

Types

Cam followers are available in the standard types shown in **Table 1**. These standard bearings have a screwdriver slot built into the stud head, while the outer ring has a cylindrical outside surface.

Bearings of the following unique specifications are available for manufacture upon request:

1. Hexagonal socket of the stud head (refer to the Bearing Table)
2. Eccentric bush fixed to the stud to make it an eccentric stud (refer to the Bearing Table)
3. An outer ring with a spherical outside surface (refer to page B179)

These specifications are indicated in the bearing number. Please use the Formulation of Bearing Numbers as indicated on the following page when placing orders.

In addition to the above, products with their exposed surfaces treated with a black oxide layer to enhance corrosion resistance, as well as products of special shapes and specifications, can also be manufactured. Please contact NSK for further details.

Table 1 Bearing Types

Code		Details
Metric	FCR	Full complement, without seal
	FCRS	Full complement, with seal, with thrust washer
	FCJ	With Cage, without seal
	FCJS	With Cage, with seal, with thrust washer
Inch	CR	Full complement
	CRS	Full complement, with seal, with thrust washer

Formulation of Bearing Numbers

Metric

FCR / FCRS / FCJ / FCJS — **B** **E** **16** **R**

- **R**: Omitted: Outer ring has a cylindrical outside surface. R: Outer ring has a spherical outside surface.
- **16**: Outside diameter of outer ring in mm
- **E**: Omitted: Without bush; E: With eccentric bush
- **B**: Omitted: Stud head with screwdriver slot; B: Stud head with hexagonal socket

(Example) **FCRE-16** Full complement, with screwdriver slot, with eccentric bush, outside diameter of outer ring is 16 mm

FCJSB-16 With cage, with seal/thrust washer, with hexagonal socket, outside diameter of outer ring is 16 mm

FCRSBE-16R Full complement, with seal/thrust washer, with hexagonal socket, with eccentric bush, outside diameter of outer ring is 16 mm and the outer ring has a spherical outside surface

Inch

CR / CRS — **B** **C** **E** **8**

- **8**: Outside diameter of outer ring in 1/16-inch units (Example) **8**=8/16 =1/2" **16**=16/16=1" **32**=32/16=2" **36**=36/16=2 1/4"
- **E**: Omitted: Without bush; E: With eccentric bush
- **C**: Omitted: Outer ring has a cylindrical outside surface. C: Outer ring has a spherical outside surface.
- **B**: Omitted: Stud head with screw driver slot; B: Stud head with hexagonal socket

(Example) **CRE-8** Full complement, with screwdriver slot, with eccentric bush, outside diameter of outer ring is 12.700 mm (8/16 inch)

CRSB-8 Full complement, with seal/thrust washer, with hexagonal socket, outside diameter of outer ring is 12.700 mm (8/16 inch)

CRSBCE-8 Full complement, with seal/thrust washer, with hexagonal socket, with eccentric bush, outside diameter of outer ring is 12.700 mm (8/16 inch) and the outer ring has a spherical outside surface

Spherical Radius of Outer Ring Outside

Spherical radii for metric bearings are shown in **Table 2** and for inch bearings in **Table 3**.

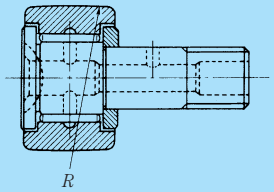


Table 2 Spherical Radius, R, of Metric Bearings

Units: mm

Bearing Number	R
All Bearing Numbers	500

Table 3 Spherical Radius, R, of Inch Bearings

Units: mm

Bearing Number (1)	R	Bearing Number (1)	R	Bearing Number (1)	R
CRC-8	150	CRC-20	360	CRC-40	760
CRC-8-1	180	CRC-22	360	CRC-44	760
CRC-10	180	CRC-24	500	CRC-48	760
CRC-10-1	200	CRC-26	500	CRC-52	760
CRC-12	250	CRC-28	500	CRC-56	760
CRC-14	250	CRC-30	500	CRC-64	760
CRC-16	300	CRC-32	600	CRC-80	1200
CRC-18	300	CRC-36	600	CRC-96	1400

Note (1) This uses CRC bearings as an example. However, since dimension series are the same, the R dimensions shown in the above table should also be applied to CRSC bearings.

Accuracy

NSK cam followers are manufactured with the dimensional tolerances shown in **Table 4**. Running accuracy is subject to ISO Class 0 and is always applied to the bearing before surface treatment.

Table 4 Cam Follower Dimensional Tolerances

Units: μm

Type		Stud Diameter Deviation, Δ_{dmp}, or Fitting Tolerance	Single Plane Mean Outside Diameter Deviation. Δ_{Dmp}		Deviation of Outer Ring Width C, Δ_{Cs}
			Cylindrical Outside	Spherical Outside	
Metric	FCR, FCRS, FCJ, FCJS	h7	According to ISO Class 0	0 −50	According to ISO Class 0
Inch	CR, CRS	+25 0	0 −25	0 −50	0 −130

Accessories

NSK cam followers include a sealing plug but the nut, washer and grease nipple are available as accessories. When ordering, please add the part code shown in **Table 5** after the bearing number.

Table 5 Part Code of Accessories

Type	Sealing Plug	Grease	Nut	Washer	Nipple
FCR, FCJ, CR	Included	Not pre-packed	I	N	Z
FCRS, FCJS, CRS	Included	pre-packed			

[Example] FCR-16 + I N Z

- Z — Nipple
- N — Washer
- I — Nut

Applications

Fitting

The recommended fittings for the mounting area of cam follower studs are shown in **Table 6**. Since cam followers use cantilever mounting, the fitting for the stud and mounting hole should be assembled with as little play as possible. In case there is interference in the fitting of the stud and mounting hole, the center of the stud end should be inserted with a press. The side plate should be securely fixed onto the plane surface square to the center axis, without play. In order to get a sufficient support face for the side plate, the dimension of the shoulder should be larger than the value of F shown in the bearing table. Play in the mounting part should be avoided because it would not only cause inclination to the stud but also create considerable deflection when a load is imposed. The load would also be unevenly imposed on the rollers due to edge contact between the outer ring and track (on which the outer ring runs).

Table 6 Recommended Fitting for Stud Mounting Part

Type		Fitting Tolerance of Mounting Hole
Metric	FCR, FCJ, FCRS, FCJS	JS7 (J7)
Inch	CR, CRS	F7

Maximum Permissible Load

The maximum radial load that the cam follower can carry is determined by the bearing strength and shear strength of the stud rather than the load rating for needle bearings. This value is given in the bearing table as the maximum permissible load.

The cam follower can deliver a performance of the maximum strength of stud by mounting securely to the side plate so as to minimize overhang. It is necessary, therefore, to minimize chamfering at the corner of the stud mounting hole and avoid burrs.

Permissible Track Load

The permissible load of the track is determined by compression strength or hardness. The permissible load of the track shown in the bearing table is the value of a track made of steel with a hardness of HRC40.

Table 7 indicates the permissible load coefficient of the track for each hardness.

The permissible load of the track for each hardness can be obtained by multiplying the permissible load of the track shown in the bearing table by the permissible load coefficient of the track corresponding to each hardness.

Table 7 Permissible Load Coefficient of Track

Hardness (HRC)	Coefficient
20	0.4
25	0.5
30	0.6
35	0.8
40	1.0
45	1.4
50	1.9
55	2.6
58	3.2

Maximum Clamp Torque

Since the stud of the cam follower receives bending stress and tensile stress from the bearing load, the screw clamp torque should not exceed the value shown in the bearing table. The clamp torque (maximum) in the bearing table is the value when the thread is lubricated and the value when dry should be approximately doubled.

Pre-packed Grease

The cam follower with a seal is pre-lubricated with lithium soap-based grease and the working temperature range is -10 to +110°C.

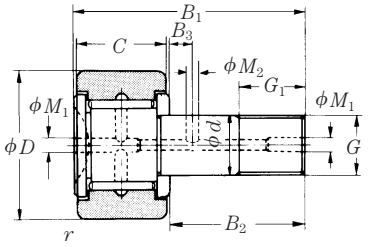
FCR (Full Complement)

FCRS (Full Complement, Sealed, With Thrust Washer)

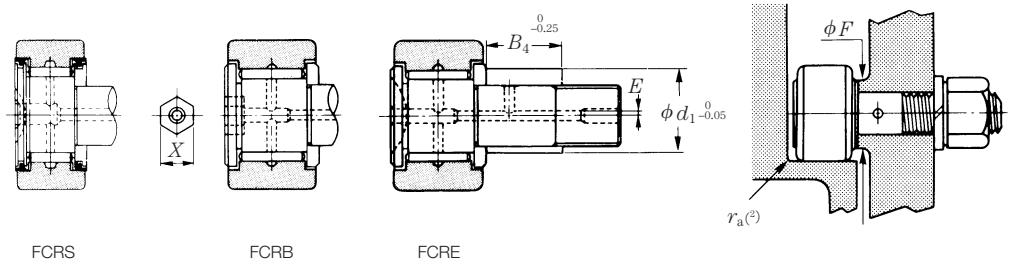
FCJ (With Cage)

FCJS (Sealed, With Cage and Thrust Washer)

Outside Diameter 10 — 62 mm



FCR



FCRS FCRB FCRE

*1 Dimensions of Hexagonal Socked (width across flats) (3)
*2 Shoulder Dimensions

Boundary Dimensions (mm)			Dimensions (mm)								Bearing Numbers	
D	C	d	Screw G	G_1	B_1	B_2	B_3	M_2	M_1	r min	FCR FCJ	FCRS FCJS
10	7	3	M3X0.5	5	17	9	—	—	—	0.3	**FCJB-10**	
12	8	4	M4X0.7	6	20	11	—	—	—	0.3	**FCJ-12**	
	8	4	M4X0.7	6	20	11	—	—	—	0.3	**FCJB-12**	
16	11	6	M6X1.0	8	28	16	—	—	4(1)	0.3	**FCR-16**	**FCRS-16**
	11	6	M6X1.0	8	28	16	—	—	4(1)	0.3	**FCJ-16**	**FCJS-16**
19	11	8	M8X1.25	10	32	20	—	—	4(1)	0.3	**FCR-19**	**FCRS-19**
	11	8	M8X1.25	10	32	20	—	—	4(1)	0.3	**FCJ-19**	**FCJS-19**
22	12	10	M10X1.25	12	36	23	—	—	4(1)	0.3	**FCR-22**	**FCRS-22**
	12	10	M10X1.25	12	36	23	—	—	4(1)	0.3	**FCJ-22**	**FCJS-22**
26	12	10	M10X1.25	12	36	23	—	—	4(1)	0.3	**FCR-26**	**FCRS-26**
	12	10	M10X1.25	12	36	23	—	—	4(1)	0.3	**FCJ-26**	**FCJS-26**
30	14	12	M12X1.5	13	40	25	6	3	6	0.6	**FCR-30**	**FCRS-30**
	14	12	M12X1.5	13	40	25	6	3	6	0.6	**FCJ-30**	**FCJS-30**
32	14	12	M12X1.5	13	40	25	6	3	6	0.6	**FCR-32**	**FCRS-32**
	14	12	M12X1.5	13	40	25	6	3	6	0.6	**FCJ-32**	**FCJS-32**
35	18	16	M16X1.5	17	52	32.5	8	3	6	0.6	**FCR-35**	**FCRS-35**
	18	16	M16X1.5	17	52	32.5	8	3	6	0.6	**FCJ-35**	**FCJS-35**
40	20	18	M18X1.5	19	58	36.5	8	3	6	1	**FCR-40**	**FCRS-40**
	20	18	M18X1.5	19	58	36.5	8	3	6	1	**FCJ-40**	**FCJS-40**
47	24	20	M20X1.5	21	66	40.5	9	4	8	1	**FCR-47**	**FCRS-47**
	24	20	M20X1.5	21	66	40.5	9	4	8	1	**FCJ-47**	**FCJS-47**
52	24	20	M20X1.5	21	66	40.5	9	4	8	1	**FCR-52**	**FCRS-52**
	24	20	M20X1.5	21	66	40.5	9	4	8	1	**FCJ-52**	**FCJS-52**
62	29	24	M24X1.5	25	80	49.5	11	4	8	1	**FCR-62**	**FCRS-62**
	29	24	M24X1.5	25	80	49.5	11	4	8	1	**FCJ-62**	**FCJS-62**

Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Track Loads (N)	{kgf}	Mass (kg)	*1 (mm)	Eccentric Bush Dimensions (mm) (4)			*2 (mm)	Tightening Torque (5) (N·cm)	{kgf·cm}
C_r		P_{max}				approx	X	B_4	d_1	E	F min	max	max
1 390	142	590	60	1 320	135	0.005	2.5	—	—	—	7.5	28	2.9
1 970	201	1 050	107	1 860	190	0.008	—	—	—	—	9	64	6.5
1 970	201	1 050	107	1 860	190	0.008	2.5	—	—	—	9	64	6.5
5 800	590	2 360	240	3 350	340	0.020	4	8	9	0.5	11	226	23
2 830	288	2 360	240	3 350	340	0.018	4	8	9	0.5	11	226	23
6 600	670	4 200	425	4 150	425	0.031	4	10	11	0.5	13	550	56
3 450	355	4 200	425	4 150	425	0.030	4	10	11	0.5	13	550	56
8 550	875	6 550	665	5 300	540	0.047	5	11	13	0.5	15	1 060	108
4 350	445	6 550	665	5 300	540	0.045	5	11	13	0.5	15	1 060	108
8 550	875	6 550	665	6 000	610	0.060	5	11	13	0.5	15	1 060	108
4 350	445	6 550	665	6 000	610	0.058	5	11	13	0.5	15	1 060	108
12 500	1 280	9 250	945	7 800	795	0.088	6	12	17	1	20	1 450	148
7 200	735	9 250	945	7 800	795	0.086	6	12	17	1	20	1 450	148
12 500	1 280	9 250	945	8 050	820	0.099	6	12	17	1	20	1 450	148
7 200	735	9 250	945	8 050	820	0.096	6	12	17	1	20	1 450	148
18 600	1 900	17 000	1 740	11 800	1 200	0.17	10	15.5	22	1	24	4 000	410
9 700	990	17 000	1 740	11 800	1 200	0.165	10	15.5	22	1	24	4 000	410
20 500	2 090	21 700	2 220	14 300	1 460	0.25	10	17.5	24	1	26	5 950	605
10 300	1 050	21 700	2 220	14 300	1 460	0.24	10	17.5	24	1	26	5 950	605
28 200	2 880	26 400	2 690	20 800	2 120	0.39	12	19.5	27	1	31	8 450	860
19 200	1 950	26 400	2 690	20 800	2 120	0.38	12	19.5	27	1	31	8 450	860
28 200	2 880	26 400	2 690	22 900	2 340	0.47	12	19.5	27	1	31	8 450	860
19 200	1 950	26 400	2 690	22 900	2 340	0.455	12	19.5	27	1	31	8 450	860
40 000	4 100	38 500	3 950	34 000	3 450	0.80	14	24.5	34	1	45	15 200	1 550
24 900	2 540	38 500	3 950	34 000	3 450	0.79	14	24.5	34	1	45	15 200	1 550

Note (1) Only the head of the stud has an oil hole.
Remarks Standard grease is packed in sealed cam followers, but not in cam followers without seals.

Notes (2) Should not be greater than $r_{(min)}$.
(3) Applicable to FCRB only.
(4) Applicable to FCRE only.
(5) These values are for when the screw is oiled; they should be approximately doubled when the screw is dry.

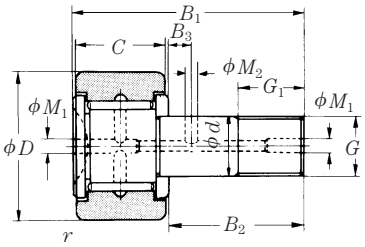
FCR (Full Complement)

FCRS (Full Complement, Sealed, With Thrust Washer)

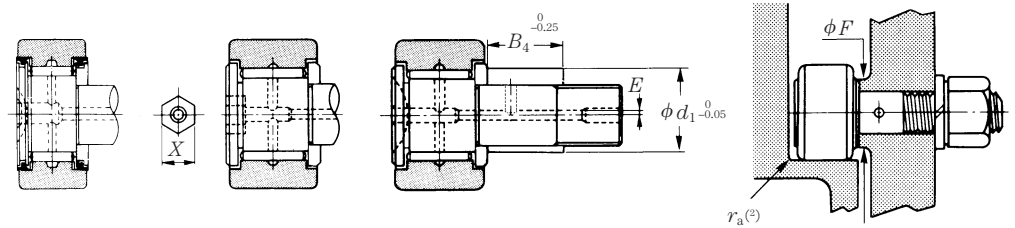
FCJ (With Cage)

FCJS (Sealed, With Cage and Thrust Washer)

Outside Diameter 72 — 90 mm



FCR



FCRS FCRB FCRE

*1 Dimensions of Hexagonal Socked (width across flats) (3)
*2 Shoulder Dimensions

Boundary Dimensions (mm)			Dimensions (mm)								Bearing Numbers	
D	C	d	Screw G	G_1	B_1	B_2	B_3	M_2	M_1	r min	FCR FCJ	FCRS FCJS
72	29	24	M24X1.5	25	80	49.5	11	4	8	1	**FCR-72**	**FCRS-72**
	29	24	M24X1.5	25	80	49.5	11	4	8	1	**FCJ-72**	**FCJS-72**
80	35	30	M30X1.5	32	100	63	15	4	8	1	**FCR-80**	**FCRS-80**
	35	30	M30X1.5	32	100	63	15	4	8	1	**FCJ-80**	**FCJS-80**
85	35	30	M30X1.5	32	100	63	15	4	8	1	**FCR-85**	**FCRS-85**
	35	30	M30X1.5	32	100	63	15	4	8	1	**FCJ-85**	**FCJS-85**
90	35	30	M30X1.5	32	100	63	15	4	8	1	**FCR-90**	**FCRS-90**
	35	30	M30X1.5	32	100	63	15	4	8	1	**FCJ-90**	**FCJS-90**

Basic Dynamic Load Ratings (N)	{kgf}	Limiting Loads (N)	{kgf}	Limiting Track Loads (N)	{kgf}	Mass (kg)	*1 (mm)	Eccentric Bush Dimensions (mm) (4)			*2 (mm)	Tightening Torque (5) (N·cm)	{kgf·cm}
C_r		P_{max}				approx	X	B_4	d_1	E	F min	max	max
40 000	4 100	38 500	3 950	38 000	3 860	1.05	14	24.5	34	1	45	15 200	1 550
24 900	2 540	38 500	3 950	38 000	3 860	1.05	14	24.5	34	1	45	15 200	1 550
60 500	6 200	61 000	6 200	52 000	5 300	1.55	17	31	40	1.5	52	30 500	3 120
39 000	4 000	61 000	6 200	52 000	5 300	1.55	17	31	40	1.5	52	30 500	3 120
60 500	6 200	61 000	6 200	55 500	5 650	1.75	17	31	40	1.5	52	30 500	3 120
39 000	4 000	61 000	6 200	55 500	5 650	1.75	17	31	40	1.5	52	30 500	3 120
60 500	6 200	61 000	6 200	59 000	6 000	1.95	17	31	40	1.5	52	30 500	3 120
39 000	4 000	61 000	6 200	59 000	6 000	1.95	17	31	40	1.5	52	30 500	3 120

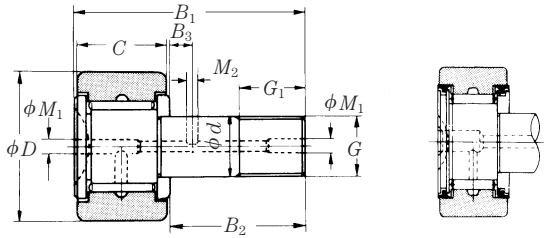
Remarks Standard grease is packed in sealed cam followers, but not in cam followers without seals.

Notes (2) Should not be greater than $r_{(min)}$.
(3) Applicable to FCRB only.
(4) Applicable to FCRE only.
(5) These values are for when the screw is oiled; they should be approximately doubled when the screw is dry.

CR (Full Complement)

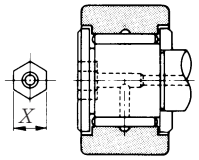
CRS (Full Complement, Sealed, With Thrust Washer)

Outside Diameter 12.700 — 152.4 mm

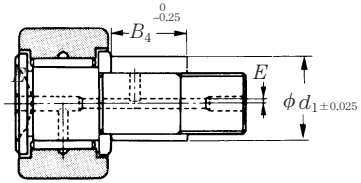


CR

CRS



CRB



CRE

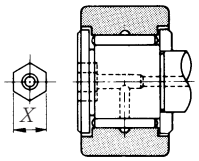
Boundary Dimensions (mm, inch) D		C	d		Screw UNF G	G_1	B_1	B_2	B_3	M_2	M_1
12.700	0.5000	8.74	4.826	—	10-32	6.4	22.8	12.70	—	—	3.2(3)
		9.52	4.826	—	10-32	6.4	26.8	15.88	—	—	3.2(3)
15.875	0.6250	10.31	6.350	0.2500	1/4-28	7.9	27.3	15.88	—	—	3.2(3)
		11.13	6.350	0.2500	1/4-28	7.9	31.3	19.05	—	—	3.2(3)
19.050	0.7500	12.70	9.525	0.3750	3/8-24	9.5	36.1	22.22	6.35	2.4	4.8
22.225	0.8750	12.70	9.525	0.3750	3/8-24	9.5	36.1	22.22	6.35	2.4	4.8
25.400	1.0000	15.88	11.112	0.4375	7/16-20	12.7	42.4	25.40	6.35	3.2	4.8
28.575	1.1250	15.88	11.112	0.4375	7/16-20	12.7	42.4	25.40	6.35	3.2	4.8
31.750	1.2500	19.05	12.700	0.5000	1/2-20	15.9	52.2	31.75	7.94	3.2	4.8
34.925	1.3750	19.05	12.700	0.5000	1/2-20	15.9	52.2	31.75	7.94	3.2	4.8
38.100	1.5000	22.22	15.875	0.6250	5/8-18	19.0	61.6	38.10	9.52	4.0	4.8
41.275	1.6250	22.22	15.875	0.6250	5/8-18	19.0	61.6	38.10	9.52	4.0	4.8
44.450	1.7500	25.40	19.050	0.7500	3/4-16	22.2	71.2	44.45	11.11	4.0	4.8
47.625	1.8750	25.40	19.050	0.7500	3/4-16	22.2	71.2	44.45	11.11	4.0	4.8
50.800	2.0000	31.75	22.225	0.8750	7/8-14	25.4	83.9	50.80	12.70	4.8	4.8
57.150	2.2500	31.75	22.225	0.8750	7/8-14	25.4	83.9	50.80	12.70	4.8	4.8
63.500	2.5000	38.10	25.400	1.0000	1-14(1)	28.6	96.9	57.15	14.29	4.8	4.8
69.850	2.7500	38.10	25.400	1.0000	1-14(1)	28.6	96.9	57.15	14.29	4.8	4.8
76.200	3.0000	44.45	31.750	1.2500	1 1/4-12	31.8	109.6	63.50	15.88	4.8	6.4
82.550	3.2500	44.45	31.750	1.2500	1 1/4-12	31.8	109.6	63.50	15.88	4.8	6.4
88.900	3.5000	50.80	34.925	1.3750	1 3/8-12	34.9	122.3	69.85	17.46	4.8	6.4
101.600	4.0000	57.15	38.100	1.5000	1 1/2-12	38.1	147.7	88.90	19.05	4.8	6.4
127.000	5.0000	69.85	50.800	2.0000	2-12(2)	65.1	200.9	128.59	22.22	4.8	6.4
152.400	6.0000	82.55	63.500	2.5000	2 1/2-12(2)	76.2	237.5	152.40	25.40	4.8	6.4

Bearing Numbers CR	CRS	Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}	Limiting Track Loads (N)	{kgf}
CR-8	**CRS-8**	3 800	385	1 340	136	2 050	209
CR-8-1	**CRS-8-1**	4 200	430	1 340	136	2 260	230
CR-10	**CRS-10**	5 350	545	2 640	269	2 920	298
CR-10-1	**CRS-10-1**	5 800	595	2 640	269	3 200	325
CR-12	**CRS-12**	8 850	900	5 900	600	4 550	465
CR-14	**CRS-14**	8 850	900	5 900	600	5 150	525
CR-16	**CRS-16**	11 200	1 140	6 950	710	6 550	670
CR-18	**CRS-18**	11 200	1 140	6 950	710	7 450	760
CR-20	**CRS-20**	18 500	1 890	9 600	975	10 400	1 060
CR-22	**CRS-22**	18 500	1 890	9 600	975	11 500	1 170
CR-24	**CRS-24**	21 700	2 220	15 900	1 620	15 000	1 530
CR-26	**CRS-26**	21 700	2 220	15 900	1 620	16 300	1 660
CR-28	**CRS-28**	27 700	2 830	23 600	2 400	20 000	2 040
CR-30	**CRS-30**	27 700	2 830	23 600	2 400	21 500	2 190
CR-32	**CRS-32**	41 500	4 250	32 500	3 350	29 100	2 970
CR-36	**CRS-36**	41 500	4 250	32 500	3 350	33 000	3 350
CR-40	**CRS-40**	52 500	5 350	43 000	4 400	41 000	4 200
CR-44	**CRS-44**	52 500	5 350	43 000	4 400	45 500	4 650
CR-48	**CRS-48**	76 000	7 750	67 000	6 850	60 000	6 100
CR-52	**CRS-52**	76 000	7 750	67 000	6 850	64 500	6 600
CR-56	**CRS-56**	99 000	10 100	81 500	8 350	81 500	8 300
CR-64	**CRS-64**	126 000	12 800	98 000	9 950	107 000	10 900
CRB-80	**CRSB-80**	198 000	20 200	176 000	18 000	154 000	15 700
CRB-96	**CRSB-96**	283 000	28 800	277 000	28 200	226 000	23 000

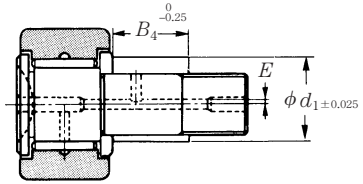
Notes (1) USN screws are used in place of UNF screws.
(2) UN screws are used in place of UNF screws.
(3) Only the head of the stud has an oil hole.

Remarks Standard grease is packed in sealed cam followers, but not in cam followers without seals.

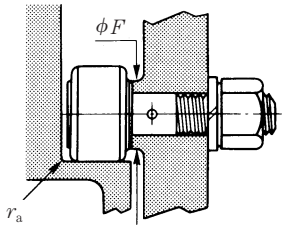
Abutment and fillet dimensions, and tightening torque are shown on the next page.



CRB



CRE



*1 Dimensions of Hexagonal Socked (width across flats) (4)

Boundary Dimensions CR	Boundary Dimensions CRS	Mass (kg) approx	*1 (mm) X	Eccentric Bush Dimensions (mm) (5) B_4	d_1	E
CR-8	**CRS-8**	0.010	3.18	—	—	—
CR-8-1	**CRS-8-1**	0.010	3.18	9.52	6.350	0.25
CR-10	**CRS-10**	0.019	3.18	—	—	—
CR-10-1	**CRS-10-1**	0.020	3.18	11.10	9.525	0.38
CR-12	**CRS-12**	0.034	4.76	12.70	12.700	0.38
CR-14	**CRS-14**	0.044	4.76	12.70	12.700	0.38
CR-16	**CRS-16**	0.073	6.35	12.70	15.875	0.76
CR-18	**CRS-18**	0.089	6.35	12.70	15.875	0.76
CR-20	**CRS-20**	0.135	6.35	15.88	17.450	0.76
CR-22	**CRS-22**	0.16	6.35	15.88	17.450	0.76
CR-24	**CRS-24**	0.24	7.94	19.05	22.225	0.76
CR-26	**CRS-26**	0.275	7.94	19.05	22.225	0.76
CR-28	**CRS-28**	0.385	7.94	22.22	25.400	0.76
CR-30	**CRS-30**	0.43	7.94	22.22	25.400	0.76
CR-32	**CRS-32**	0.62	11.11	25.40	30.150	0.76
CR-36	**CRS-36**	0.755	11.11	25.40	30.150	0.76
CR-40	**CRS-40**	1.15	12.70	28.58	34.925	0.76
CR-44	**CRS-44**	1.35	12.70	28.58	34.925	0.76
CR-48	**CRS-48**	1.9	15.88	31.75	44.450	1.52
CR-52	**CRS-52**	2.2	15.88	31.75	44.450	1.52
CR-56	**CRS-56**	2.9	15.88	34.92	46.025	1.52
CR-64	**CRS-64**	4.3	19.05	50.80	50.800	1.52
CRB-80	**CRSB-80**	8.7	22.22	—	—	—
CRB-96	**CRSB-96**	15	25.40	—	—	—

Notes (4) Applicable to CRB only.
(5) Applicable to CRE only.

*2 Fillet Radius of Housing

Shoulder Dimensions (mm) F min	*2 (mm) r_a min	Tightening Torque (6) (N·cm) max	{kgf·cm} max
7.5	0.25	90	9.2
7.5	0.25	90	9.2
9.1	0.4	226	23
9.1	0.4	226	23
12.7	0.4	630	64
12.7	0.4	630	64
15.1	0.8	1 700	173
15.1	0.8	1 700	173
19.0	0.8	2 310	236
19.0	0.8	2 310	236
22.6	0.8	4 400	450
22.6	0.8	4 400	450
26.6	1	8 500	865
26.6	1	8 500	865
30.6	1.3	10 200	1040
30.6	1.3	10 200	1040
34.9	2.3	15 300	1560
34.9	2.3	15 300	1560
44.4	2.3	23 100	2360
44.4	2.3	23 100	2360
48.8	2.3	28 200	2880
57.9	2.3	34 000	3450
82.6	4	34 000	3450
99.2	4	34 000	3450

Notes (6) These values are for when the screw is oiled; they should be approximately doubled when the screw is dry.



ROLLER FOLLOWERS (YOKE TYPE TRACK ROLLERS)

			Pages
Metric Design	Bore Diameter	5–50 mm	B194–B195
Inch Design	Bore Diameter	6.350–57.150 mm	B196–B197

The roller follower with a thick-wall outer ring and crowned roller has a large load capacity which enables it to carry shock loads. Handling is easy since it it non-separating and applications such as track rollers, cams and rocker arms are wide.

Types

NSK roller rollowers are available in the types shown in **Table 1**. All of the bearings in **Table 1** have a cylindrical outside. Bearings of the following specifications are available and can be manufactured upon request:

1. Outer ring with a spherical exterior
In this case, your order should be placed in the following manner:
(1) For metric bearings, **R** should be added to the end of bearing number.
Examples: **FYCR-5R**, **FYCJ-5R**
FYCRS-5R, **FYCJS-5R**
(2) For inch bearings, **C** should be added to the end of the bearing type code.
Examples: **YCRC-12**
YCRSC-12
Spherical radius of outer ring is shown in page B193.
2. Exposed surface treated with a black oxide layer
In addition to the above, products of special shape and specifications are available and can be manufactured upon order. Please contact NSK for further details.

Table 1 Bearing Types

Type		Details
Metric	FYCR	Full complement
	FYCRS	With seal, full complement, with thrust washer
	FYCJ	With cage
	FYCJS	With seal, with cage, with thrust washer
Inch	YCR	Full complement
	YCRS	With seal, full complement, with thrust washer

Accuracy

NSK metric roller followers are manufactured in compliance with ISO Class 0 (refer to pages A26 and A27), and inch followers in compliance with the tolerances shown in **Table 2**. Running accuracy complies with ISO Class 0 in both cases.

The deviations for the above apply to the bearings before surface treatment.

Table 2 Tolerances for Inch Bearings

Units: μm

Nominal Bore Diameter, d (mm)		Single Plane Mean Bore Diameter Deviation, Δ_{dmp}		Single Plane Mean Outside Diameter Deviation, Δ_{Dmp}				Outer Ring Width Deviation, Δ_{Cs}		Overall Inner Ring Width Deviation, Δ_{Bs}	
				Cylindrical Outside Surface		Spherical Outside Surface (1)					
over	incl	high	low	high	low	high	low	high	low	high	low
—	19.05	+5	−10	0	−25	0	−50	0	−130	+130	−256
19.05	57.15	+2	−13	0	−25	0	−50	0	−130	+130	−250

Note (1) Metric bearings have the same dimension tolerance.

Applications

Fitting

Roller followers are generally used with a transition or loose fit for the shaft because their outer ring rotates. With heavy loads, however, it is recommended to harden the shaft and use it with a tight fit. In order to prevent the bearing from moving in the axial direction, both ends of the inner ring should be securely fixed to the plane surface, square to the center axis. In order to sufficiently support the side plate, the dimension of the shoulder should be larger than the value of F in the bearing table.

Recommended fittings are shown in **Table 3**.

Table 3 Recommended Shaft Fittings

Load	Fitting Tolerance of Shaft
Light Load/Normal Load	g6 or h6
Heavy Load	k6

Permissible Track Load

Permissible track load values are the same as those for cam followers. Please refer to the explanation on permissible track load on page B181.

Pre-Packed Grease

Roller followers with seals are pre-lubricated with lithium soap-based grease and the working temperature range is -10 to +110°C.

Spherical Radius of Outer Ring Outside

Spherical radii for metric bearings are shown in **Table 4** and for inch bearings in **Table 5**.

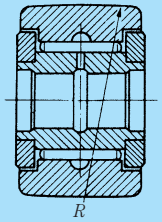


Table 4 Spherical Radius, R, of Metric Bearing Outside

Units: mm

Bearing Number	R
All Bearing Numbers	500

Table 5 Spherical Radius, R, of Inch Bearing Outside

Units: mm

Bearing Number (1)	R	Bearing Number (1)	R
YCRC-12	250	YCRC-32	600
YCRC-14	250	YCRC-36	600
YCRC-16	300	YCRC-40	760
YCRC-18	300	YCRC-44	760
YCRC-20	360	YCRC-48	760
YCRC-22	360	YCRC-52	760
YCRC-24	500	YCRC-56	760
YCRC-26	500	YCRC-64	760
YCRC-28	500	YCRC-80	1200
YCRC-30	500	YCRC-96	1400

Note (1) While YCRC bearing numbers are listed here, the values of R also apply to YCRSC bearings.

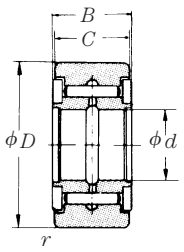
FYCR (Full Complement)

FYCRS (Full Complement, Sealed, With Thrust Washer)

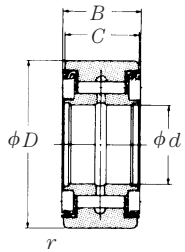
FYCJ (With Cage)

FYCJS (Sealed, With Cage and Thrust Washer)

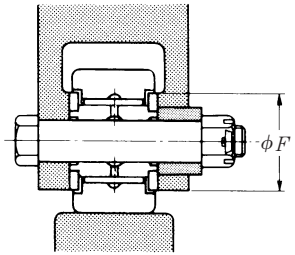
Bore Diameter 5 — 50 mm



FYCR



FYCRS



Boundary Dimensions (mm)					Basic Load Ratings (N)		{kgf}		Limiting Track Loads (N)	{kgf}
d	D	C	B $^{0}_{-0.38}$	r min	C_r	C_{or}	C_r	C_{or}		
5	16	11	12	0.3	5 800	8 000	590	815	3 350	340
	16	11	12	0.3	2 830	2 620	288	267	3 350	340
6	19	11	12	0.3	6 550	9 900	665	1 010	4 150	425
	19	11	12	0.3	3 450	3 600	355	365	4 150	425
8	24	14	15	0.3	10 100	15 000	1 030	1 530	6 500	665
	24	14	15	0.3	5 700	6 000	580	610	6 500	665
10	30	14	15	0.6	11 700	18 500	1 190	1 890	7 800	795
	30	14	15	0.6	6 950	8 200	705	835	7 800	795
12	32	14	15	0.6	12 600	21 000	1 280	2 140	8 050	820
	32	14	15	0.6	7 650	9 650	780	985	8 050	820
15	35	18	19	0.6	18 700	29 300	1 910	2 990	11 800	1 200
	35	18	19	0.6	12 200	14 100	1 250	1 440	11 800	1 200
17	40	20	21	1	21 100	35 000	2 160	3 600	14 300	1 460
	40	20	21	1	13 700	16 700	1 390	1 700	14 300	1 460
20	47	24	25	1	28 900	50 000	2 940	5 100	20 800	2 120
	47	24	25	1	18 200	22 600	1 850	2 310	20 800	2 120
25	52	24	25	1	32 500	60 000	3 300	6 100	22 900	2 340
	52	24	25	1	22 200	31 000	2 270	3 150	22 900	2 340
30	62	28	29	1	47 500	96 000	4 800	9 800	33 000	3 350
	62	28	29	1	31 500	47 000	3 200	4 800	33 000	3 350
35	72	28	29	1	49 500	106 000	5 050	10 800	36 500	3 700
	72	28	29	1	33 000	52 500	3 400	5 350	36 500	3 700
40	80	30	32	1	54 500	126 000	5 600	12 800	43 500	4 450
	80	30	32	1	38 500	67 500	3 950	6 900	43 500	4 450
45	85	30	32	1	57 500	139 000	5 850	14 100	46 500	4 750
	85	30	32	1	40 000	73 000	4 100	7 450	46 500	4 750
50	90	30	32	1	60 500	152 000	6 150	15 500	49 500	5 050
	90	30	32	1	41 500	78 000	4 200	7 950	49 500	5 050

Bearing Numbers		Mass (kg)	Shoulder Dimensions (mm)
FYCR FYCJ	FYCRS FYCJS	approx	F min
FYCR-5	**FYCRS-5**	0.016	10
FYCJ-5	**FYCJS-5**	0.014	10
FYCR-6	**FYCRS-6**	0.022	12
FYCJ-6	**FYCJS-6**	0.020	12
FYCR-8	**FYCRS-8**	0.044	14
FYCJ-8	**FYCJS-8**	0.042	14
FYCR-10	**FYCRS-10**	0.069	17
FYCJ-10	**FYCJS-10**	0.067	17
FYCR-12	**FYCRS-12**	0.076	19
FYCJ-12	**FYCJS-12**	0.074	19
FYCR-15	**FYCRS-15**	0.105	23
FYCJ-15	**FYCJS-15**	0.097	23
FYCR-17	**FYCRS-17**	0.145	25
FYCJ-17	**FYCJS-17**	0.14	25
FYCR-20	**FYCRS-20**	0.255	29
FYCJ-20	**FYCJS-20**	0.245	29
FYCR-25	**FYCRS-25**	0.285	34
FYCJ-25	**FYCJS-25**	0.275	34
FYCR-30	**FYCRS-30**	0.48	51
FYCJ-30	**FYCJS-30**	0.47	51
FYCR-35	**FYCRS-35**	0.64	58
FYCJ-35	**FYCJS-35**	0.635	58
FYCR-40	**FYCRS-40**	0.88	66
FYCJ-40	**FYCJS-40**	0.865	66
FYCR-45	**FYCRS-45**	0.93	72
FYCJ-45	**FYCJS-45**	0.91	72
FYCR-50	**FYCRS-50**	0.995	76
FYCJ-50	**FYCJS-50**	0.965	76

Remarks Standard grease is packed in sealed cam followers, but not in cam followers without seals.